

GIT18

:tada:

Use for the first commit in a new repository.
:tada: start the project

:bug:

Correct wrong behaviour in RTL, testbench, firmware, model, or flow code.
:bug: gemm: fix the tile count for odd partial tiles

:ambulance:

Repair something that blocks the whole team.
:ambulance: widen the counters for larger workloads

:sparkles:

Add a new function that is complete and operational.
:sparkles: dw: add stride-2 depthwise support

:hatching_chick:

Record the first time that something operates from end to end.
:hatching_chick: dual-core hello world runs on hardware

:construction:

Commit work that is not complete.
:construction: layernorm: first draft of the datapath

:boom:

Change an interface that other users depend on.
:boom: csr: change the numbers of the control register fields

:recycle:

Change the structure of the code, but do not change its behaviour.
:recycle: move the normalization datapath out of the output path

:art:

Change names, layout, or format only.
:art: align the serializer with the stream fifo style

:fire:

Delete files that are no longer necessary.
:fire: delete the old golden models

:coffin:

Delete code that no other code uses.
:coffin: delete the unused normquant load state

:wastebasket:

Mark code, a register, or a mode for future removal.
:wastebasket: deprecate the old identity normquant mode

:truck:

Move or rename files, modules, or directories.
:truck: move the register interface into sw/

:rewind:

Undo a previous commit.
:rewind: revert the pad injector change

:twisted_rightwards_arrows:

Merge one branch into another.
:twisted_rightwards_arrows: merge main into devel

:adhesive_bandage:

Make a small correction that is not critical.
:adhesive_bandage: add a guard around the assertion for synthesis

:see_no_evil:

Add or update the `.gitignore`` file.
:see_no_evil: ignore the generated waveforms

:bookmark:

Add a release tag, or change the version number of the repository.
:bookmark: increment the version for the next patch release

RTL13

:electric_plug:

Connect blocks that already exist.
:electric_plug: connect clear to all engine submodules

:abacus:

Change how the design computes.
:abacus: make the accumulator 40 bits wide

:vertical_traffic_light:

Control when data moves and which requester wins.
:vertical_traffic_light: change the backup mode to round-robin arbitration

:dna:

Make a fixed value configurable.
:dna: parameterize the input serializer for other instances

:control_knobs:

Change the control and status registers or the register interface.
:control_knobs: regenerate the register interface for the new stride field

:bridge_at_night:

Cross a clock domain or a reset domain.
:bridge_at_night: uart: synchronize rx_valid into the core clock

:shield:

Add or change protection against faults.
:shield: interco: propagate the ecc bits through the router

:space_invader:

Change correct RTL because a tool cannot process it.
:space_invader: flatten the interface array ready for elaboration

:fire_extinguisher:

Repair a defect that can stay in the silicon.
:fire_extinguisher: break the combinational loop in the enable path

:compass:

Change how the design calculates or arranges addresses.
:compass: addressgen: add zig-zag tile traversal

:satellite_antenna:

Change an off-chip link.
:satellite_antenna: eth: handle the pause frame backoff

:magnet:

Work on emulation or on the laboratory setup.
:magnet: fpga: add the board target with ddr4 and pcie

:label:

Add or change type definitions.
:label: add an fsm phase type to the package

TIMING2

:scissors:

Change the RTL to meet the timing target.
:scissors: register the ready signal out of the memory

:triangular_ruler:

Change the constraints instead of the RTL.
:triangular_ruler: set the io delays to 0.5*TCK and add path groups

PPA4

:zap:

Decrease the number of cycles or increase the throughput.
:zap: overlap the compute of the next tile with the current store

:chart_with_downwards_trend:

Decrease the number of gates, flip-flops, memories, LUTs, or FIFO entries.
:chart_with_downwards_trend: decrease the sink fifo from 8 to 2 entries

:battery:

Decrease the power that the design uses.
:battery: gate the clock of the lanes that are not active

:stopwatch:

Measure the performance, but do not change it.
:stopwatch: tb: count the cycles of each load and store phase

BACKEND2

:factory:

Change the implementation flow.
:factory: add gate-level simulation to the flow

:cityscape:

Change the physical layout.
:cityscape: update the floorplan for the new memory macros

VERIFICATION7

:white_check_mark:

Add, extend, or repair tests and regressions.
:white_check_mark: add partial tiling tests with stalls

:petri_dish:

Change the environment that runs the tests.
:petri_dish: tb: add a timeout so that failed tests stop early

:game_die:

Change how you make the test data.
:game_die: generate depthwise kernels as test specifications

:spider_web:

Collect or increase the coverage.
:spider_web: exclude the testbench core from the coverage report

:microscope:

Make the internal behaviour visible.
:microscope: add a transaction tracer to the interconnect

:safety_vest:

Add or change the checks that the design must satisfy.
:safety_vest: assert that the bank count is a power of two

:rotating_light:

Correct the warnings that a tool gives.
:rotating_light: correct the width truncation warnings

FIRMWARE2

:boot:

Change the code that starts and supports the system.
:boot: initialize the core structures at startup

:joystick:

Change the code that operates the hardware.
:joystick: hal: add the stride-2 depthwise configuration

MODELING3

:mirror:

Change the reference model that you compare the hardware against.
:mirror: model the exponent packing to match the rtl

:monocle_face:

Examine the results.
:monocle_face: plot the memory occupancy against the fsm state

:snake:

Change Python code that is not the model and not an analysis.
:snake: divide utils.py into separate modules

BUILD5

:hammer:

Change the machinery that builds and runs the project.
:hammer: runner: rebuild the rtl only when it changes

:wrench:

Change the settings, not the machinery.
:wrench: sim: increase the default testbench memory

:construction_worker:

Add or change the continuous integration setup.
:construction_worker: ci: divide the build and test stages

:green_heart:

Repair a pipeline that fails.
:green_heart: ci: correct the cache directory

:bricks:

Change the infrastructure around the flow.
:bricks: add a cache for the regression artifacts

DEPENDENCIES5

:arrow_up:

Change the version of an IP, a submodule, a tool, or a package.
:arrow_up: bump the control ip for autotrigger support

:pushpin:

Change which revision you use, but not which version you want.
:pushpin: point the dependency at the open-source release

:heavy_plus_sign:

Add an IP, a submodule, a package, or a tool.
:heavy_plus_sign: add matplotlib for the analysis scripts

:heavy_minus_sign:

Remove an IP, a submodule, a package, or a tool.
:heavy_minus_sign: remove the unused yaml parser

:alien:

Adapt to a change that you do not control.
:alien: parse the new utilization report format

DOCS4

:memo:

Add or update the documentation.
:memo: document the supported output modes

:bulb:

Add or update the comments in the source code.
:bulb: explain why this signal is registered

:pencil2:

Correct a spelling error in the documentation, the comments, or a message.
:pencil2: correct the register field name in the readme

:page_facing_up:

Add or update the license and the file headers.
:page_facing_up: add the license headers to the rtl sources